TECHNICAL DESIGN DOCUMENT · RELEASE REVIEW

Travel operations, made more coherent.

A technical account of Itera's package publishing, enquiry handling, AI assistance, access controls and verified release boundaries.

LIVE APPLICATION
https://itera-5nq4.vercel.app/

REVIEWED RELEASE
4c58cbb · master

PREPARED
09 November 2026

PRODUCT SLICE
Core build + selected optional depth

This document distinguishes implemented code, production-verified behavior, mocked integrations and items that exist only in the uncommitted workspace. It does not treat the broader optional feature list as mandatory scope.

01 · PRODUCT AND SCOPE

One platform for the enquiry journey

Itera is a full-stack travel platform for a Kerala and South India tour operator. Its core purpose is to let staff publish and maintain package content without developer intervention, while helping the team capture, triage and answer traveller enquiries from a shared workflow.

Scope: required core versus optional depth

Scope class	Included product capabilities	Release status
Core build	Public package listing/detail; package CMS and history; enquiry form; staff inbox and detail; sign-in/registration and roles; AI extraction, matching and reply draft; seed data; mocked external messaging.	**Implemented** with the production AI verification caveat documented in §6.
Optional depth	Price estimator, map, blog CMS/SEO, analytics, activity history, branded quotes/PDF, privacy requests, scheduled publishing, sitemap, robots and legacy redirects.	Implemented selectively; features and constraints are described individually below.
Out of scope / mocked	Live payments, hotel/flight inventory, real email and WhatsApp delivery, native applications, multilingual publishing and production booking fulfilment.	Not represented as working integrations.

> **Design goal.** Preserve a clear human decision point: the original enquiry remains visible, generated data is stored separately, uncertain extraction is surfaced for review, and staff compose or edit the response.

What a visitor or staff member can do

Traveller

Search published trips; compare itinerary, hotels, price, inclusions and exclusions; submit a consented enquiry; create an account; and review owned enquiries.

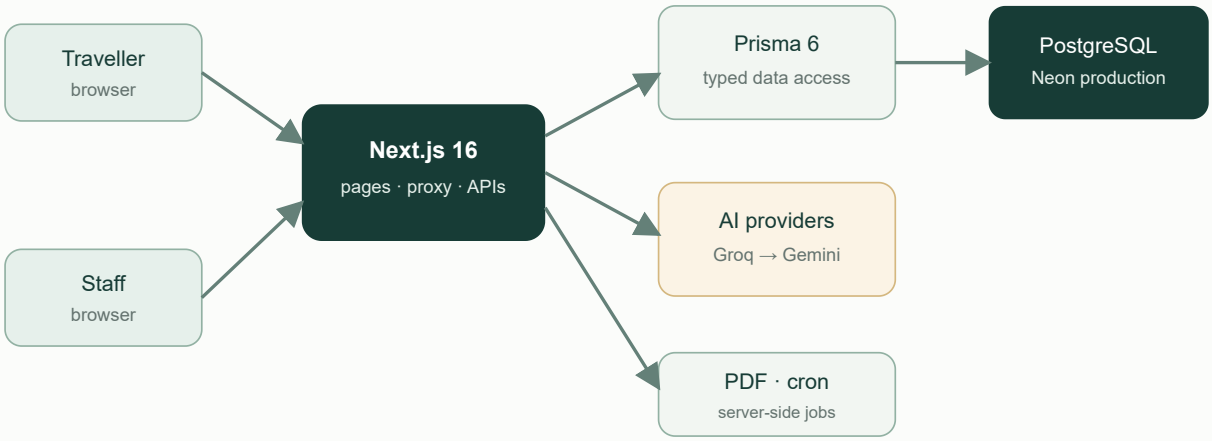
Staff

Draft and publish packages, review version history, triage enquiries, correct AI fields, add notes, assign work, prepare replies and create package-based quotes.

02 · ARCHITECTURE

Server-rendered application, relational core

A single Next.js application serves public and staff routes, API handlers and scheduled tasks. PostgreSQL stores the operational record; provider and document work is performed from server-side routes.



Browser requests enter through Next.js. The proxy handles session gates, redirects and response security headers; route handlers still perform authorization and validation before mutations.

Technology decisions

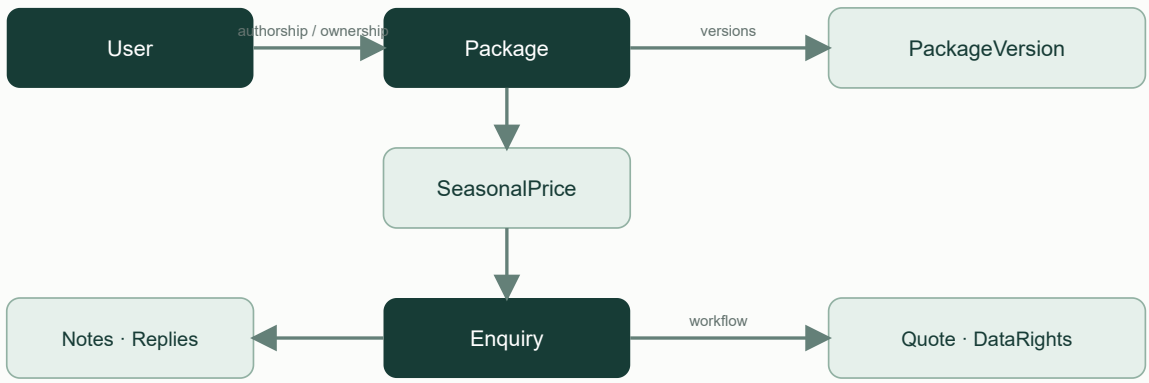
Technology	Reason for this domain	Trade-off
Next.js App Router + React + TypeScript	SEO-friendly public routes and server-side rendering share deployment and routing with APIs.	Framework-specific server/client boundaries require careful component placement.
PostgreSQL + Prisma	Packages, versions, prices, enquiries and assignments have relational constraints; transactions keep publish state coherent.	Hosted database availability and connection limits are operational dependencies.
NextAuth credentials + bcrypt	Supports seeded demo accounts and separate staff/traveller role claims without a separate auth service.	Credentials are demo-only; OAuth/SSO is not configured.
Groq + Gemini, behind server routes	Provides two extraction options; application matching and validation remain deterministic.	Provider output is variable; production field-level verification is incomplete.
Tailwind, Leaflet, pdf-lib, Vitest	Responsive UI, no-key map tiles, server-side PDF output and focused test feedback.	External tile/image hosts and PDF output still need production monitoring.

Runtime stack verified from package.json, next.config.ts, prisma/schema.prisma and src/app route structure.

03 · DATA ARCHITECTURE

Stable identities, versioned content, traceable enquiries

Package identity is deliberately separate from editable package content. The stable package row owns the slug and publication pointers; each content revision is an append-only version row. Seasonal prices remain relational so search and date-based selection can query active prices directly.



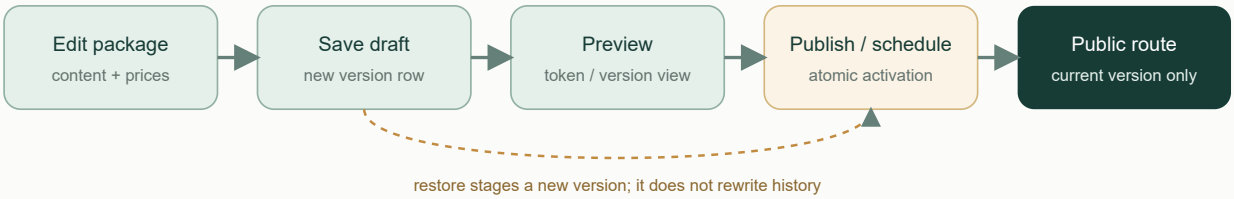
One User can submit/own enquiries and author versions, replies, notes, quotes and activity records. Enquiry may optionally reference a package and links to notes, replies, quotes and data-rights requests.

Model	Responsibility and key relationships
User	Unique email, bcrypt password hash and role. Relates to submitted/assigned enquiries and authored staff activity.
Package / PackageVersion	Stable slug, status and current/draft version pointers; version holds itinerary JSON, content, hotels, photos and map data.
SeasonalPrice	Per-person price, month interval, basis, GST treatment and active flag; linked to Package.
Enquiry	Original contact/message, dates, party/budget, channel, consent provenance, status, assignee and AI fields stored separately.
EnquiryNote / EnquiryReply	Staff-authored operational history linked to an enquiry and author.
Quote	Package/enquiry reference, token, travellers, date, server-selected pricing, GST, validity and author.
BlogPost / PreviewCache	Published editorial content and expiring token-backed package previews.
DataRightsRequest / ActivityLog	Privacy request status and administrative accountability records.

Why JSON is used selectively. Package itinerary and AI extraction have nested or evolving structures. Stable identities, dates, roles, publication state and seasonal prices remain typed relational fields for filtering and integrity.

04 · PACKAGE CMS

Draft, preview, publish—with the live version protected



The package id and URL remain stable while content evolves. A scheduled draft keeps the current published content live until publication executes.

Authoring and pricing

Package versions contain title, destination, duration, description, inclusions/exclusions, hotel tier/names, day-by-day itinerary, photo URLs and alt text, map centre, and a snapshot of seasonal prices. Seasonal prices are also stored as related rows with start/end month, per-person price, basis, GST treatment and active state.

The editor supports create/edit, duplicate, preview, publish now, scheduled publishing, history and restore. Restoring copies an earlier snapshot into a new version number. Staged content and inactive price rows are activated transactionally so the public route does not show a new itinerary with old prices.

Validation and ownership

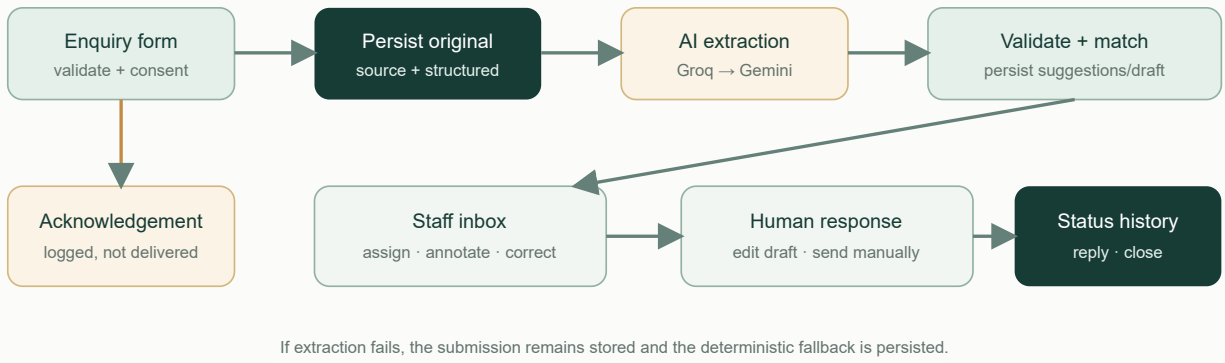
- Zod validates package forms and seasonal-price rows before writes.
- Role checks restrict edit, restore and publication actions; restore and immediate publication are Admin-only.
- Published pages query the current version; draft preview uses a short-lived preview token.
- Price selection is based on travel month; inclusive and additional GST are calculated from stored pricing parameters.

> **Operational constraint.** Vercel schedules publishing once daily at 03:00 UTC. A scheduled edit can be delayed until the next run. The slug is package-level and cannot be staged as part of a version change.

Source areas: src/app/admin/packages, src/app/api/admin/packages, src/lib/package-publishing.ts, src/lib/seasonal-pricing.ts and prisma/schema.prisma.

05 · ENQUIRY LIFECYCLE

Keep source input, extraction and staff action distinct



If extraction fails, the submission remains stored and the deterministic fallback is persisted.

Intake and inbox

`POST /api/enquiries` rate-limits and parses JSON, validates with Zod, requires privacy consent, stores the original message and structured fields, logs a fixed acknowledgement template, then calls the secret-protected internal AI route before returning. If extraction is unavailable or fails, the caller logs the failure and the internal route has a fallback; enquiry intake is not replaced by a success-shaped provider response.

The staff inbox supports status/assignee and creation-date filters plus search. The detail screen places source text next to extracted fields, recommendations, consent state, notes and the editable reply. Consultant access is scoped to assigned enquiries in protected screens/routes.

Workflow states

State	Operational meaning
NEW	Captured and awaiting staff review.
IN_PROGRESS	Assigned/being reviewed; staff notes and follow-up are available.
REPLIED	A staff response is recorded; email delivery is not integrated.
CLOSED	Work is closed; eligible records are subject to configured retention processing.

06 · AI BEHAVIOR AND EVIDENCE

Model extraction; deterministic application decisions

AI is used for interpreting free-text travel requests, not for authority over price, availability or booking. The internal route extracts dates, destination, adults/children, budget, requirements and confidence; the application validates the structure, overlays structured form values where available, and ranks published packages using application logic.

Provider, parsing and fallback

1. Try Groq when `GROQ_API_KEY` is configured; its request asks for JSON output.
2. Parse provider text, validate against `aiExtractedSchema`, and fail provider parsing if malformed or out of schema.
3. If Groq fails and Gemini is configured, try Gemini. A regression test covers malformed Groq content followed by valid Gemini output and persisted destination/suggestion data.
4. If providers are unavailable/fail, use form fields, selected package context and conservative deterministic fallback; suggestions can be empty.
5. Compute confidence in application code. Lower-confidence results are assigned for consultant review where a consultant is available, and the draft explicitly avoids confirming an itinerary/price.

Recommendation and reply policy

Suggestions are scored from published package destination, trip duration, budget against active seasonal prices, season and itinerary requirement tokens. Results are ordered and capped at three. The model does not set the price: current seasonal price rows provide price context. Draft language changes at confidence thresholds (0.40 and 0.75) and labels preliminary or uncertain details.

The generated reply is stored in `Enquiry.aiDraftReply`. Staff may correct fields and edit the reply. The code does not dispatch that reply automatically; actual email/WhatsApp integration is mocked. The fixed acknowledgement is logged server-side rather than sent by an email provider.

> **Production evidence: partial.** A controlled synthetic production enquiry returned HTTP 200 and was persisted in Neon. Travel dates, adults/children, budget and draft reply were present; destination was null and package suggestions were empty. Thus production persistence is verified for those fields only—not complete extraction or recommendations. Malformed JSON was verified to return HTTP 400.

Groq and Gemini keys were confirmed configured in Vercel Production without disclosing values. Historical runtime/request logs were unavailable within the available Hobby-plan retention window, so the exact provider response behind the null destination/empty matches could not be established. Code-level failover improvements and regression tests do not prove provider behavior in production.

AI quote release boundary

The deployed release can create an indicative quote on the high-confidence path and include tokenized page/PDF links in the stored draft. Staff must review the draft reply before sending, but the deployed release does not enforce a separate approval check on that AI-generated quote token before public page/PDF access.

A quote-approval gate and related tests exist only in the current uncommitted workspace and are not in deployed commit `4c58cbb`. Therefore approval-gated public access for AI-generated quotes is **not verified in the deployed release**.

> **Human oversight is not a substitute for provider verification.** The correct operational response to missing/uncertain extraction is consultant review, not treating the fallback as a confirmed recommendation.

Evidence: src/app/api/enquiries/route.ts; src/app/api/internal/ai/route.ts; tests/enquiry-submission-route.test.ts; tests/internal-ai-extraction-route.test.ts; production synthetic enquiry verification recorded 2026-11-09.

PRODUCTION VERIFICATION RECORD

What the controlled production run established

Synthetic enquiry; secrets and customer data excluded.

Check	Observed evidence	Status
Provider keys	Groq and Gemini were configured in Vercel; values were not exposed.	Confirmed
Submission	Synthetic enquiry returned HTTP 200 and was persisted in Neon.	Verified
Persisted fields	Dates, adult/child counts, budget and draft reply were present.	Verified
Destination	The persisted extracted destination was null.	Incomplete
Suggestions	The production record contained zero package suggestions.	Incomplete
Malformed JSON	Malformed enquiry body returned HTTP 400.	Verified
AI quote gate	The deployed commit has no approval check before token page/PDF access.	Not verified

Diagnosis remains unresolved

Historical Vercel runtime/request logs were unavailable within the available retention window. Configured keys do not prove successful extraction; destination and recommendation behavior require another controlled production verification.

07 · IDENTITY, ACCESS AND PRIVACY

Role boundaries plus route-level checks

Role	Intended access in the application	Examples
ADMIN	Broad staff administration.	Package publish/restore, enquiries, blog, quotes, analytics, activity and data-rights queue.
EDITOR	Editorial content workflows.	Blog creation and editing; not general enquiry or package administration.
CONSULTANT	Assigned enquiry and package operations.	Assigned inbox, package workflow, quotes and analytics as authorized by route.
TRAVELLER	Own account and submitted enquiries.	Public packages, registration/login, own enquiry status and privacy request.

Implemented safeguards

- bcrypt password hashes and NextAuth JWT session strategy.
- Proxy session gate plus independent checks inside protected pages/APIs.
- Zod validation and parameterized Prisma operations for common mutation paths.
- Internal AI secret and cron bearer secret; cron routes fail closed without configuration.
- IP-based rate limits on login/enquiry and selected public mutation routes.
- Consent timestamp, source and version stored with enquiry; separate WhatsApp consent/withdrawal fields.
- Correction/deletion request workflow with Admin resolution and transactional updates.
- Content Security Policy and security response headers in `src/proxy.ts`.

Privacy lifecycle and release boundary

The shipped retention route selects closed enquiries older than `ENQUIRY_RETENTION_DAYS` (default 730) and anonymizes parent-enquiry personal fields. Correction/deletion request handling also updates linked data transactionally.

> **Related records.** In deployed commit `4c58cbb`, scheduled retention does not anonymize the content of related notes, replies and quotes. A broader linked-record anonymization implementation and tests exist in the uncommitted workspace only; they have not been shipped or production-verified.

Operational and audit history should be preserved without retaining traveller identifiers where possible. Retention extensions should be deployed only after review and tests confirm which quote financial/package fields must remain for legitimate history.

Operational caveats

- The rate limiter uses process memory; multiple serverless instances do not share counters.
- Demo seed passwords are intentionally public and must never be used for real accounts.
- Redirects are permanent (301), but the client's original 140 URL list was not supplied; exact mapping is unverified.

- External image and OpenStreetMap tile hosts are allowed by CSP; broad all-route browser console verification is not claimed.

08 · EXPERIENCE AND APPLICATION SURFACE

Editorial public pages; task-focused staff screens

The interface uses warm ivory and stone neutrals, a restrained forest-green accent, editorial serif headings and compact sans-serif utility text. Public package pages foreground photography, clear itinerary/pricing information and the enquiry call to action. Staff screens prioritize status, assignment and editable business data over marketing presentation.

Screen	Design decision	Workflow supported
Home / packages	Responsive travel imagery, search filters, cards and progressive content loading.	Discover and narrow package choices on mobile.
Package detail	Itinerary, gallery/map, seasonal prices, estimator and enquiry form in one route.	Evaluate trip details and ask about a specific package.
Package editor	Form fields for day-by-day content, photos/alt text, hotels and seasonal rates; preview and history actions.	Make controlled updates without modifying live content directly.
Enquiry inbox/detail	Status/assignee filters, response-time indicators and source/AI/staff content presented together.	Prioritize, assign, correct, annotate and respond.
Blog CMS	Visual editing affordances with Markdown-compatible persistence and SEO fields.	Publish travel content while keeping the existing renderer format.
Quote/PDF	Branded quote representation with itemized GST and validity.	Share a package-based indicative price; approval boundary is qualified in §6.

Key API areas

Area	Representative route
Public package data	`GET /api/packages`; `GET /api/packages/[slug]`
Enquiry lifecycle	`POST /api/enquiries`; `/api/admin/enquiries/*`
Package CMS	`/api/admin/packages/*`
AI extraction	`POST /api/internal/ai` (internal secret required)
Authentication	`/api/auth/[...nextauth]`; `/api/auth/register`
Privacy/scheduled work	`/api/me/data-rights`; `/api/admin/data-rights`; `/api/cron/*`
Quote rendering	`/quote/[token]`; `/api/quotes/[token]/pdf`

09 · VERIFICATION

Evidence, not implied success

Local checks

Check	Result in reviewed workspace	Qualification
Vitest	**18 files / 114 tests passed**	Current workspace includes uncommitted application/test changes not in the deployed commit.
TypeScript	**npx tsc --noEmit passed**	Current workspace.
Production build	**npm run build passed**	Next.js 16.4.5 / Turbopack; routes generated successfully.
ESLint	**0 errors**, 19 warnings	Warnings include existing image and navigation recommendations; not warning-free.

Deployed production Lighthouse · mobile Slow 4G

Measured on the deployed application with Lighthouse 13.6.0; devtools throttling, 150 ms RTT, 1638.4 Kbps throughput, 4× CPU slowdown, 412×823 mobile viewport.

Route	Baseline LCP	After LCP	FCP after	TBT after	CLS after	Transfer after
Homepage	5.735 s	**4.953 s**	2.337 s	171 ms	0.000	920,593 B
Packages	3.144 s	**2.518 s**	2.185 s	231 ms	0.000	829,667 B
Munnar detail	4.566 s	**3.765 s**	3.099 s	265 ms	0.000	994,376 B

Target status. All three routes improved. Packages narrowly missed a 2.5-second LCP target by 18 ms; homepage and Munnar detail remain above target. These are production measurements, separate from earlier local-build results.

Other production evidence and limits

- The deployed commit `4c58cbbeb3db0e9e42aa53358b2a5420bca935cf` reached Vercel deployment success; homepage, packages listing and the seeded Munnar detail URL returned HTTP 200.
- One synthetic production enquiry confirmed HTTP 200, storage, dates, traveller counts, budget and draft reply; destination and suggestions were missing as noted in §6.
- Malformed enquiry JSON returned HTTP 400 after the regression fix.
- Redirect generation was structurally checked (140 package-pattern mappings plus static aliases in code). The client's original 140 URL inventory is unavailable, so exact correspondence and destination correctness are not independently verified.

10 · ASSUMPTIONS AND NEXT ENGINEERING WORK

Explicit assumptions; measured next steps

Type	Decision / qualification
Assignment boundary	The required scope is the Core Build section of the brief. Smart Touches and other extras are optional depth, not retroactively mandatory acceptance criteria.
Hosting	Vercel hosts the Next.js application; PostgreSQL is hosted on Neon for the live demo.
Pricing	Seeded prices are INR, per person and typically twin-sharing. GST treatment and rate are stored/configurable; actual quotation remains indicative until staff confirms availability.
Media	Package/blog photography is stored as URLs, not uploaded through an asset-management workflow.
Response time	The UI communicates a response-time expectation; no SLA worker or live reminder service enforces it.
Language	English-first. Hindi/German product localization is not implemented; prompts may understand informal/Hinglish input but this is not localization.
Integrations	External email, WhatsApp, payments and live inventory are mocked or out of scope as permitted for the demo.

Prioritized follow-up

1. **Resolve and re-verify production AI extraction.** Collect a fresh controlled synthetic enquiry and permitted provider logs; verify destination and recommendations are persisted before claiming complete AI behavior.
2. **Ship and test release-boundary safety work.** Review/commit the local AI-quote approval and linked-record retention changes; then verify public page/PDF access and idempotent retention against a disposable database.
3. **Continue measured LCP work.** Address the measured hero resource load duration, then re-run the same deployed profile rather than assuming the local result applies.
4. **Obtain the client's original redirect inventory.** Compare every supplied source/destination pair and test permanent redirect behavior against it.
5. **Replace process-local rate limiting** with a shared service before relying on enforcement across serverless instances.

> **Submission boundary.** The reviewed deployment and production measurements refer to pushed SHA `4c58cbb`. Additional quote-approval and related-record retention work visible in the local workspace is not part of that deployed release.

Prepared from the repository implementation, Prisma schema/seed, route tests, deployment checks and production Lighthouse evidence. No live secret values or customer records are included.

Companion documents · Repository · Live application · Deployment guide